

Highlights

Random Matrix Filtering and Planar Financial Networks for Volatility Forecasting in the Brazilian Stock Market

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- RMT separates market, group and noisy modes in Brazilian equities.
- PMFG filtering reveals localized sectoral dependency structures.
- Within-sector correlations exceed between-sector correlations.
- Network features add incremental evidence for volatility forecasting.
- GARCH and HAR-RV set strong benchmarks for network-based ML.

Random Matrix Filtering and Planar Financial Networks for Volatility Forecasting in the Brazilian Stock Market

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ABSTRACT

Financial markets are complex systems in which collective fluctuations, sectoral organization and noisy microscopic dependencies coexist in empirical correlation matrices. This paper develops an econophysics pipeline for Brazilian equities traded on B3 from 1998 to 2025 and evaluates whether the resulting dependency structure contains information for volatility forecasting. Using adjusted daily prices, we compute log returns for a historical universe of 58 equities and combine stylized-fact diagnostics, Random Matrix Theory (RMT), hierarchical clustering, Minimum Spanning Trees (MSTs), Planar Maximally Filtered Graphs (PMFGs) and realized-volatility forecasting models. RMT separates a dominant market mode, group/sector modes and a residual noisy component; five empirical eigenvalues lie above the Marcenko–Pastur upper bound, and the largest eigenvalue accounts for about 37% of the spectrum. Filtered MST and PMFG representations show that removing the market mode reduces average edge correlations while preserving localized sectoral organization. The group-mode PMFG has lower mean edge correlation but higher clustering than the original network, suggesting a more modular mesoscopic structure. Forecasting experiments for PETR4, VALE3 and BBDC4 indicate that GARCH(1,1) and HAR-RV remain strong benchmarks, especially at short horizons, while machine-learning models with PMFG-derived features provide competitive and incremental evidence rather than uniformly dominating classical models. The results support the view that RMT-filtered topology is useful for interpreting market structure and may complement volatility-risk modelling in emerging equity markets.

1. Introduction

Financial markets are complex adaptive systems. Their observed price dynamics are generated by heterogeneous investors, institutional constraints, liquidity cycles, information flows and feedback mechanisms that operate at several time scales. Even when the empirical object is reduced to a panel of daily returns, the market is not a collection of independent univariate processes. It contains collective motion, sectoral organization, idiosyncratic shocks, crisis amplification and persistent volatility. A central task of empirical econophysics is to identify which part of this structure is robust enough to be interpreted and which part is dominated by noise.

Dependence is particularly important because financial risk is inherently cross-sectional. The volatility of one asset matters, but portfolio risk, contagion, diversification and systemic vulnerability depend on how assets move together. During calm periods, correlations may appear moderate and dispersed. During crises, common movement often increases, and the effective dimensionality of the market can collapse. A method that only studies isolated return series misses this collective component. Conversely, a method that only studies the full sample correlation matrix may confuse persistent structure with sampling noise.

Emerging equity markets make this problem more interesting. They combine global risk channels with local institutional and macroeconomic features. Brazil is a useful case because B3 includes globally exposed commodity companies, large financial institutions, regulated utilities, consumer cyclical, industrial firms, telecommunications

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firms and real-estate companies. These assets share exposure to domestic monetary policy, exchange-rate dynamics and country risk, but they also respond to distinct sectoral fundamentals. The resulting dependency matrix is therefore expected to contain a dominant market mode, sectoral modes and a large residual component. Despite these attractive properties, the Brazilian equity market remains underexplored in the complex-systems literature. While studies have addressed Brazilian interbank networks [36], global financial system linkages [33] and agrarian cross-correlations [35], a comprehensive econophysics pipeline combining RMT filtering, PMFG construction and volatility forecasting has not been systematically applied to a broad B3 equity universe.

The empirical difficulty is that correlation matrices are noisy. When the number of assets is not negligible relative to the number of observations, a nontrivial part of the eigenspectrum can be compatible with random sampling variation. Random Matrix Theory (RMT) provides a benchmark for this problem by comparing empirical eigenvalues with the spectrum expected from a random correlation matrix [24, 19, 29]. Eigenvalues outside the random band are not automatically economic facts, but they identify modes that deserve interpretation. This makes RMT a useful intermediate layer between raw correlations and financial network construction.

Network filtering provides a complementary representation. The Minimum Spanning Tree (MST) converts correlations into a sparse connected backbone with $N - 1$ edges [22]. The Planar Maximally Filtered Graph (PMFG) retains a richer planar structure with $3N - 6$ edges and preserves triangles and 4-cliques [37]. These networks allow the researcher to ask whether the strongest dependencies are organized by sectors, whether hubs change after market-mode removal and whether the mesoscopic topology contains interpretable information.

This paper uses Brazilian equities to connect three empirical layers. The first layer documents stylized facts, static correlations, sectoral correlation differences and rolling synchronization. The second layer applies RMT to separate market, group/sector and noise components, then studies clustering and filtered matrices. The third layer converts these matrices into MST and PMFG networks and evaluates whether the resulting topological features add information to realized-volatility forecasting.

The forecasting extension is intentionally conservative. Classical volatility models such as GARCH(1,1) [4] and HAR-RV [11] are strong benchmarks because volatility is persistent. A network-based forecasting model should not be expected to replace this persistence. The more realistic question is whether topology adds incremental information when combined with classical predictors. This paper therefore treats machine learning as an empirical test of the informational content of RMT-filtered networks, not as the main conceptual contribution.

The contribution of this paper is fourfold:

1. It builds a reproducible econophysics pipeline for B3 equities using adjusted daily prices from 1998 to 2025, including return construction, correlation diagnostics, sectoral comparison and rolling synchronization.
2. It applies RMT to a 58-asset historical core universe and separates the correlation matrix into market, group/sector and noise components.
3. It compares MST and PMFG networks built from original and RMT-filtered matrices, with emphasis on hub reorganization, clique structure, clustering and subsector aggregation.
4. It evaluates whether RMT- and PMFG-derived features add incremental predictive information to realized-volatility forecasting relative to GARCH(1,1), HAR-RV and classical volatility predictors, using a nested ablation design across three feature sets.

The article is organized as follows. Section 2 reviews financial dependencies, RMT, filtered financial networks and volatility forecasting. Section 3 describes the data and asset universe. Section 4 presents the empirical pipeline. Sections 5–10 report the results. Sections 11 and 12 discuss interpretation and limitations, and Section 13 concludes.

2. Related work

2.1. Financial dependencies and econophysics

Econophysics studies financial markets using tools from statistical physics, nonlinear dynamics, complex systems and network science. A central empirical observation is that financial returns display stylized facts that are difficult to reconcile with independent Gaussian models. Heavy tails, volatility clustering, slow decay in absolute-return autocorrelations and collective market modes are recurring findings across markets and time scales [23, 10]. These patterns motivate empirical methods that focus on distributions, dependence and topology rather than only on mean returns.

Financial dependency analysis sits between econophysics and financial economics. Raddant and Di Matteo [32] emphasize that correlations, causality measures, volatility spillovers and networks provide complementary views of market structure. The present paper follows this perspective. The empirical goal is not simply to estimate a correlation matrix, but to decompose it, filter it, transform it into networks and test whether the extracted structure has forecasting relevance.

Cross-correlation dynamics during crises have received particular attention. Zheng et al. [41] showed that changes in the cross-correlation structure can serve as early indicators of systemic instability. Billio et al. [3] proposed econometric measures of connectedness and systemic risk in financial sectors. Junior and Franca [17] documented how correlations among global stock indices increase during crises, reducing the effective dimensionality of the market. These findings motivate the rolling-correlation and crisis-synchronization analysis undertaken in this paper.

2.2. Random Matrix Theory in financial correlations

RMT entered empirical finance as a way to separate information from noise in large correlation matrices. Laloux et al. [19, 20] showed that a large part of the eigenspectrum of financial correlation matrices can resemble the spectrum of random matrices described by the Marčenko–Pastur law [24]. Plerou et al. [29] further studied eigenvalues and eigenvectors outside the random band and interpreted the leading eigenvalue as a market-wide mode. These studies provide the foundation for the spectral decomposition used here.

Subsequent work extended RMT-based cleaning methods for practical applications. Bun et al. [8] reviewed and unified several approaches for cleaning large covariance matrices using random matrix theory, providing a comprehensive toolkit for practitioners and researchers. The mathematical foundations of random matrix theory are covered in the review by Guhr et al. [14].

The RMT interpretation used in this manuscript is deliberately cautious. The largest eigenvalue is interpreted as the Market Mode because it captures broad positive co-movement across many assets. Eigenvalues above the Marčenko–Pastur upper edge, excluding the first one, are interpreted as group or sector candidates. Eigenvalues inside the random band are treated as noisy residual components. This terminology is a filtering convention, not a claim that every eigenvector has a unique economic label.

2.3. Financial networks: MST, PMFG and filtered graphs

Mantegna [22] introduced the MST as a way to reveal hierarchical organization in financial markets. The method converts correlations into distances and retains the connected tree with minimum total distance. Because the MST has only $N - 1$ edges, it is extremely sparse. This sparsity is useful for visualization, but it also discards cycles and many local dependencies.

Subsequent studies applied the MST framework to study the dynamics and taxonomy of equity markets. Onnela et al. [26] analysed the dynamics of MST-based market correlations and their implications for portfolio analysis. Bonanno et al. [5] extended the network approach to multiple world markets, showing that MSTs consistently organize assets by sector and geography. Song et al. [34] studied the evolution of worldwide stock markets through correlation-based graphs, documenting how network structures change through time.

The PMFG relaxes the MST constraint while preserving planarity [37]. For N nodes, it retains $3N - 6$ edges and contains the MST as a subgraph when the same ranking of distances is used. The PMFG can preserve triangles and 4-cliques, making it more suitable for detecting local communities, sectoral clusters and hub reorganization. Tumminello et al. [38] further explored the relationship between correlation, hierarchies and networks in financial markets. More recently, Massara et al. [25] introduced the Triangulated Maximally Filtered Graph (TMFG), which extends the PMFG framework and can be computed more efficiently for very large networks.

Filtered networks are particularly important for risk analysis. Pozzi et al. [31] used PMFG-based analysis to show that peripheral assets in financial networks can offer better diversification than central hubs. Aste et al. [2] studied correlation dynamics in volatile markets using filtered network approaches. Kenett et al. [18] used partial correlation analysis to reveal the dominating role of the financial sector in stock market networks. These studies motivate the hub-reorganization analysis presented in this paper, where we compare how centrality rankings change after RMT-based market-mode removal.

2.4. Financial networks in emerging markets

The application of network-based methods to emerging equity markets has attracted growing interest. Tabak et al. [36] studied the topology of the Brazilian interbank network, documenting its scale-free properties and vulnerability to

targeted failures. Silva et al. [33] analysed the structure and dynamics of the global financial network, with particular attention to how emerging economies connect to the broader system. Wang et al. [39] studied the correlation structure of world stock markets using Pearson and partial correlation-based networks, showing that emerging markets display distinctive dependency patterns.

Studies focused specifically on Brazilian markets include Pereira et al. [28], who applied detrended cross-correlation analysis to construct multiscale networks of stock markets including B3, and Leonel Caetano and Yoneyama [21], who applied complex network analysis directly to the Brazilian stock market. Stosic et al. [35] studied correlations and cross-correlations in Brazilian agrarian commodities and stocks. These studies demonstrate that Brazilian financial data present interesting structural properties, but a comprehensive pipeline combining RMT filtering, PMFG construction and volatility forecasting has not been systematically applied to a broad B3 equity universe.

2.5. Volatility forecasting and machine learning

Volatility forecasting has a long econometric tradition. GARCH-type models, introduced by Engle [12] and generalized by Bollerslev [4], capture conditional heteroskedasticity through parsimonious dynamic equations. Hansen and Lunde [15] showed that the simple GARCH(1,1) is notoriously difficult to outperform systematically. HAR-RV models [11] approximate long-memory behaviour in realized volatility through daily, weekly and monthly components. The theoretical foundations of realized-volatility measurement were established by Andersen et al. [1]. Poon and Granger [30] provided an influential review of volatility forecasting methods. Forecast evaluation is delicate because realized volatility is a proxy for latent conditional variance; QLIKE is often used because of its robustness properties under imperfect volatility proxies [27].

Machine-learning methods provide flexible nonlinear approximations. Random Forests [6], gradient boosting [13] and regularized linear models [16] can combine large feature sets containing lagged volatility, market-wide variables and network centralities. Bucci [7] applied neural networks to realized volatility forecasting, showing that flexible models can capture nonlinear volatility dynamics. Christensen et al. [9] developed a machine learning framework for volatility forecasting and demonstrated the importance of careful feature engineering. Zhang and Broadstock [40] explored how financial network topology relates to market risk from a machine learning perspective.

In financial forecasting, however, flexibility can also amplify noise. For this reason, this paper evaluates machine learning against strong baselines and treats network features as potential incremental predictors rather than as a replacement for classical volatility dynamics. The nested feature-set design follows an ablation logic: Set A measures the predictive value of classical volatility information, Set B tests the incremental value of market and RMT features, and Set C tests whether network topology adds further information beyond these baselines.

3. Data and asset universe

3.1. QuantBase/B3 data source

The empirical analysis uses daily Brazilian equity data processed through the QuantBase data infrastructure and stored in a local ClickHouse database. The main source table is `quantbase.candles_1d`. The relevant price field is the adjusted closing price `adj_close`, which incorporates the adjustment factors available in the database. The full generated price and return files cover the period from 16 March 1998 to 31 December 2025.

3.2. Asset filters and adjusted prices

The asset universe is constructed to focus on ordinary exchange-traded equity instruments. The pipeline requires `adj_close > 0`, uses instruments with `cod_bdi = 02`, and excludes securities whose specification contains DR, in order to remove BDRs. ETFs, funds and benchmark instruments are not used as nodes in the main equity network. Brazilian units and preferred/common share classes are retained when they satisfy the history and liquidity filters because they are traded instruments and can reveal share-class dependencies.

Adjusted prices are essential for long historical samples because stock splits, dividends, corporate reorganizations and unit conversions can otherwise create artificial jumps. At the same time, adjusted prices are not a guarantee that every corporate action is fully corrected. This is especially important for the modern liquid top-50 universe, where preliminary validation found several large adjusted-return events. For this reason, the modern universe is reported as generated but deferred for a separate quality-control stage.

Table 1

Summary of the empirical samples used in the working draft.

Sample	Assets	Period	Main use
Demo assets	3	2006–2025	Stylized facts and first forecasting experiment
Core historical universe	58	1998–2025	Correlation, RMT, clustering and networks
Modern liquid top 50	50	1998–2025 generated	Future modern-window robustness after QC

3.3. Core historical universe

The core historical universe contains 58 assets with sufficient data coverage for the RMT, clustering and network analysis. This universe is intentionally historical rather than purely current. It allows the paper to study a long Brazilian equity sample while keeping the cross-section large enough for network analysis and small enough for interpretable figures. The RMT window used in the main spectral analysis contains $T = 1527$ observations for $N = 58$ assets.

3.4. Demo assets

PETR4, VALE3 and BBDC4 are used as demonstration assets for stylized facts and as the first forecasting targets. They were selected because they represent large, liquid and economically distinct parts of B3: oil and gas, basic materials/mining and financials. This small panel is not meant to represent the entire market. It is used to show return behaviour, fat tails, volatility clustering and the feasibility of the forecasting pipeline before expanding the forecasting design to a larger target set.

3.5. Sector and subsector classification

The initial macro-sector classification maps each core-historical asset into a sector, subsector and segment. The macro sectors include Financials, Oil Gas and Biofuels, Basic Materials, Utilities, Industrials, Consumer Cyclical, Consumer Non-Cyclical, Real Estate and Telecommunications. Utilities and Real Estate are kept separate because their risk drivers differ substantially. Utilities are more directly linked to regulation, concessions and defensive cash flows, while Real Estate is more sensitive to credit conditions and the real-estate cycle. Within Basic Materials, the subsector field distinguishes Mining, Steel and Metallurgy, Chemicals, and Paper and Cellulose.

3.6. Data quality notes and limitations

The main data-quality issue is the distinction between genuine extreme returns and artificial jumps caused by corporate-action adjustment problems. For PETR4, VALE3 and BBDC4, the largest daily returns are consistent with known stress periods such as 2008, 2015–2016 and 2020. For the modern top-50 universe, however, some observations exceed thresholds that are too large for normal daily equity returns. The working pipeline therefore keeps extreme-return flags as a separate quality-control task and avoids using the modern universe for the main RMT and network claims in this draft.

4. Methodology

4.1. Adjusted log returns

For asset i at date t , the daily log return is computed from adjusted closing prices as

$$r_{i,t} = \log(P_{i,t}) - \log(P_{i,t-1}), \quad (1)$$

where $P_{i,t}$ is the adjusted closing price. Log returns are used because they are additive across time and standard in empirical econophysics. The pipeline exports both long-format return tables and wide return matrices. The long format is convenient for diagnostics by asset, while the wide format is required for correlation, RMT and network construction.

4.2. Descriptive statistics and stylized facts

The first empirical layer documents stylized facts in a small set of representative B3 assets. For each asset, the pipeline computes mean, standard deviation, skewness, excess kurtosis, minimum and maximum returns, empirical VaR at 1% and 5%, counts of large absolute returns and autocorrelations of $|r_t|$. These diagnostics are not the main contribution of the paper, but they establish that the Brazilian series display the usual empirical ingredients of econophysics: heavy tails, volatility clustering and persistent volatility.

4.3. Correlation matrix estimation

Let R be the $T \times N$ matrix of demeaned standardized returns. The Pearson correlation matrix is

$$C = \frac{1}{T-1} R^\top R. \quad (2)$$

For pairwise descriptive analysis, correlations are computed with available observations for each asset pair. For the RMT and network analysis, a common window is used to obtain a consistent $N \times N$ matrix. The core historical matrix contains $N = 58$ assets and is the basis for the main RMT, clustering, MST and PMFG results.

4.4. Sectoral correlation comparison

Each asset is assigned to an initial macro sector and subsector. For each pair (i, j) , the pair is classified as within-sector if both assets have the same macro sector and between-sector otherwise. The resulting empirical distributions are compared using mean, median, dispersion and a one-sided Mann–Whitney U test. This test provides an early validation that the economic classification is reflected in the empirical dependency matrix before any RMT filtering is applied.

4.5. Rolling average market correlation

Static correlations hide time variation. To measure market synchronization, the pipeline computes rolling average pairwise correlations over long windows. The statistic summarizes whether the market is moving as a single common system or as a more dispersed set of sectoral components. Elevated rolling average correlation during crises motivates the subsequent RMT decomposition because it suggests that the strength of the market mode changes over time. This step provides a natural bridge to spectral analysis: a strong time-varying common component should appear as a dominant eigenmode in the correlation matrix.

4.6. Random Matrix Theory and Marcenko–Pastur bounds

For a random correlation matrix with aspect ratio $Q = T/N$, the Marcenko–Pastur bounds are

$$\lambda_{\pm} = \sigma^2 \left(1 + \frac{1}{Q} \pm 2\sqrt{\frac{1}{Q}} \right), \quad (3)$$

where $\sigma^2 = 1$ for standardized returns. Eigenvalues inside $[\lambda_-, \lambda_+]$ are treated as broadly consistent with random noise. Eigenvalues above λ_+ are candidates for collective structure. In the core historical window, $N = 58$, $T = 1527$ and $Q \approx 26.33$.

4.7. Spectral decomposition and reconstruction

Let λ_k and u_k denote the eigenvalues and eigenvectors of C . The spectral representation is

$$C = \sum_{k=1}^N \lambda_k u_k u_k^\top. \quad (4)$$

This representation allows the matrix to be reconstructed from selected modes. The reconstruction is useful because it makes it possible to study the full matrix, the dominant market mode, the group/sector modes and the noisy component separately. Numerical reconstruction errors are reported to ensure that differences between matrices are caused by mode selection rather than numerical instability.

4.8. Market Mode, Group/Sector Mode and Noise Mode

The Market Mode is defined from the first eigenpair. It captures broad co-movement and is expected to have widespread loadings across the market. The Group/Sector Mode is reconstructed from the remaining eigenpairs above the Marcenko–Pastur upper bound. This component is expected to contain sectoral or subsectoral dependencies. The Noise Mode is reconstructed from eigenpairs inside the random band. The filtered matrix combines the informative modes and suppresses the random-band component.

4.9. Mantegna distance

Correlations are converted into distances using the Mantegna metric

$$d_{ij} = \sqrt{2(1 - \rho_{ij})}. \quad (5)$$

This transformation maps high correlations to short distances and low correlations to longer distances. The same distance geometry is then used for hierarchical clustering, ordered heatmaps, MST and PMFG construction, which makes the results comparable across methods.

4.10. Hierarchical clustering and ordered heatmaps

Hierarchical clustering is applied to the Mantegna distance matrix using average linkage. The dendrograms summarize how assets merge into clusters as the distance threshold increases. Ordered heatmaps use the dendrogram order to rearrange the correlation matrices so that block structures become visually identifiable. Comparing original, filtered and group-mode heatmaps helps distinguish broad market co-movement from localized sectoral structure.

4.11. Minimum Spanning Tree

For N nodes, the MST has $N - 1$ edges. With $N = 58$, this gives 57 edges. The MST is constructed from the distance matrix and retains the connected tree with minimum total distance. It is useful as a minimal dependency backbone, but its sparseness is also a limitation: it cannot represent cycles, triangles or multiple paths between sectors.

4.12. Planar Maximally Filtered Graph

The PMFG retains the strongest edges subject to planarity. For N nodes, a PMFG has $3N - 6$ edges; with $N = 58$, this gives 168 edges. Unlike the MST, the PMFG preserves triangles and 4-cliques. Under the same ranking of pairwise distances, the MST is contained in the PMFG as a subgraph, so the PMFG can be interpreted as a less aggressive topological filter that preserves richer local geometry. Clique counts are therefore useful construction diagnostics. In this paper, PMFGs are built for original, filtered and group-mode matrices and compared using edge correlations, sectoral edge ratios, clustering, centrality and hub rankings.

4.13. Aggregated subsector dependency network

The asset-level networks are complemented by an aggregated subsector dependency network. Asset dependencies are grouped by subsector, and cross-subsector edge weights summarize average dependency strength. This mesoscopic representation is useful because it translates a dense ticker-level structure into an economically interpretable map of dependency channels among subsectors.

4.14. Volatility forecasting dataset

For forecasting, realized volatility over horizon h is computed as

$$RV_{t,h} = \sqrt{\sum_{k=1}^h r_{t+k}^2}. \quad (6)$$

The target horizons are $h = 5$ and $h = 20$ trading days. The initial forecasting experiment uses PETR4, VALE3 and BBDC4. The sample is split temporally into training, validation and test periods to avoid random cross-sectional leakage across time.

4.15. Econometric benchmarks

The benchmark set includes historical volatility, EWMA, GARCH(1,1) and HAR-RV. The GARCH(1,1) conditional variance is

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \quad (7)$$

The GARCH models are estimated with Student- t innovations to account for the fat tails documented in the stylized-fact analysis. HAR-RV uses daily, weekly and monthly realized-volatility components to approximate long-memory behaviour [11]. These benchmarks are intentionally strong because volatility persistence is difficult to outperform [15].

4.16. Machine-learning models and feature sets

Three feature sets are used in a nested ablation design. Set A contains classical return and volatility predictors, including rolling volatilities, rolling absolute returns, lagged realized volatility, EWMA volatility and higher-moment rolling statistics. Set B adds market and RMT variables such as the largest eigenvalue, market-mode share and number of eigenvalues above the random-matrix bound. Set C adds MST and PMFG features, including degree, betweenness and clustering measures from original and group-mode networks. This nesting defines a controlled ablation: Set A measures the predictive value of classical volatility information alone, Set B tests the incremental value of market-wide and RMT features, and Set C tests whether network topology adds further information beyond these baselines. Ridge regression, Random Forest and histogram gradient boosting are used to evaluate these increasingly rich feature sets.

4.17. Evaluation metrics

Forecasts are evaluated using MAE, RMSE, out-of-sample R^2 and QLIKE. The QLIKE loss is

$$QLIKE_t = \log(\hat{\sigma}_t^2) + \frac{\sigma_t^2}{\hat{\sigma}_t^2}, \quad (8)$$

where $\hat{\sigma}_t^2$ is the predicted variance proxy and σ_t^2 is the realized proxy. QLIKE is included because it is widely used for volatility forecast comparison when realized volatility is an imperfect proxy for latent variance [27].

5. Stylized facts and correlation structure

5.1. Stylized facts of selected B3 assets

The first empirical figure introduces the behaviour of PETR4, VALE3 and BBDC4. These assets are not used as a miniature market index. They are used because they provide a compact visual entry point into the Brazilian data: a state-linked oil company, a global mining company and a large private bank. Figure 1 combines four diagnostics that are standard in econophysics: normalized prices, daily returns, the complementary cumulative distribution of absolute returns and the autocorrelation of absolute returns.

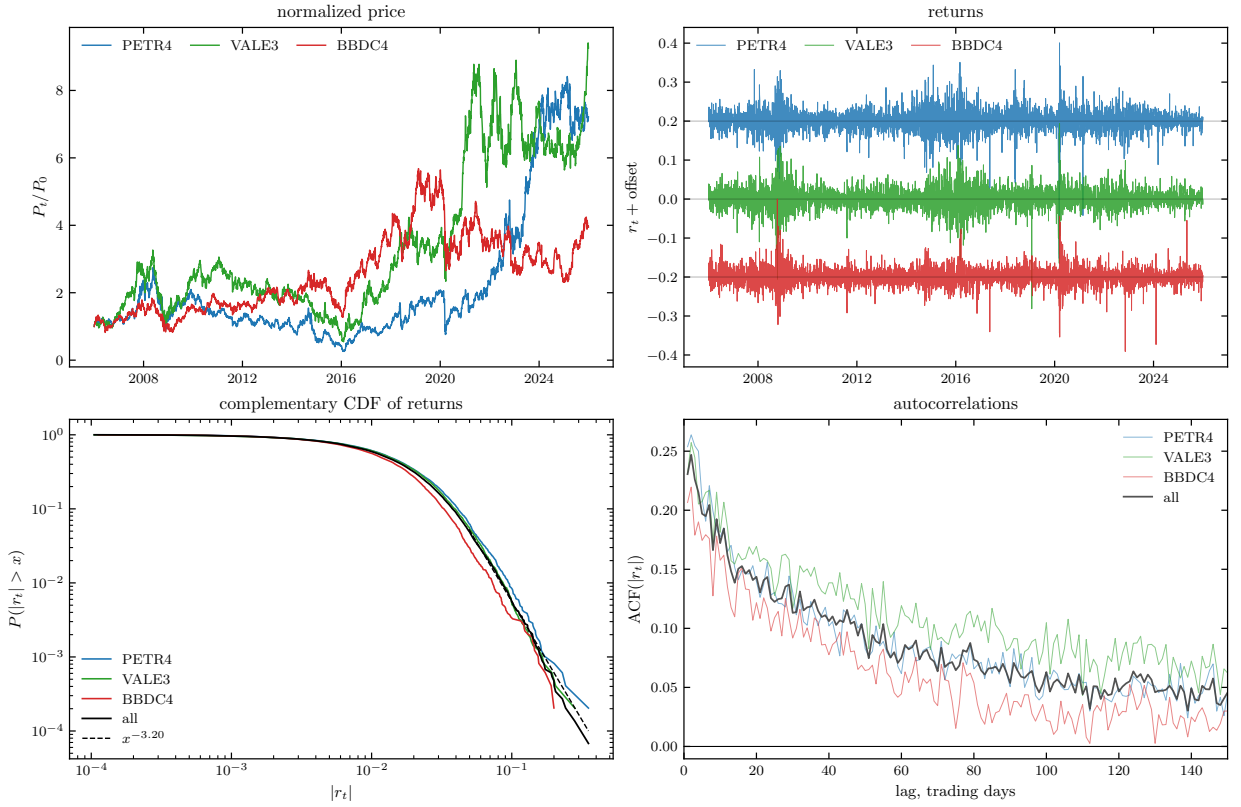


Figure 1: Stylized facts for PETR4, VALE3 and BBDC4 over 2006–2025. The panels show normalized prices, vertically offset daily log returns, the complementary cumulative distribution of absolute returns and autocorrelations of absolute returns.

The normalized-price panel shows that even a small set of liquid Brazilian assets contains heterogeneous long-run trajectories. VALE3 reflects commodity and exchange-rate cycles, PETR4 reflects oil prices and political/institutional shocks, and BBDC4 reflects financial-sector dynamics. The return panel shows volatility clustering: large shocks are not isolated uniformly through time but concentrate around stress periods. The CCDF panel shows that large absolute returns are more frequent than a Gaussian benchmark would suggest, confirming the heavy-tailed nature of B3 returns. Finally, the autocorrelation panel shows persistence in $|r_t|$, which is the empirical reason why GARCH(1,1) and HAR-RV later serve as demanding benchmarks for any volatility forecasting model.

5.2. Descriptive statistics

Table 2 quantifies the same diagnostics. All three series have means close to zero, substantial kurtosis and non-negligible counts of large absolute returns. PETR4 has the largest negative daily observation in the 2006–2025 window, while VALE3 and BBDC4 also show stress-period extremes. The autocorrelations of absolute returns remain positive across several lags, especially for VALE3.

Table 2

Descriptive statistics for the three demonstration assets, 2006–2025.

symbol	n_{obs}	mean	std	skewness	excess kurtosis	min	max	VaR 1%	VaR 5%	$ r > 10\%$	$ r > 20\%$	ACF ₁	ACF ₅	ACF ₂₀	ACF ₆₀
BBDC4	4953	0.00028	0.02180	0.02	7.96	-0.1910	0.1999	-0.0559	-0.0312	16	0	0.207	0.174	0.122	0.044
PETR4	4953	0.00041	0.02755	-0.76	11.55	-0.3524	0.2007	-0.0767	-0.0408	41	4	0.254	0.209	0.140	0.062
VALE3	4953	0.00045	0.02576	-0.22	7.51	-0.2814	0.1936	-0.0676	-0.0381	26	2	0.231	0.207	0.169	0.119

The descriptive statistics support the visual reading of Figure 1. The distributions are not symmetric Gaussian series with weak memory in volatility. They show heavy tails and persistent volatility. This justifies treating the Brazilian market as a complex empirical system rather than as a set of independent, homoskedastic return series.

5.3. Static correlation distribution

The next step moves from univariate behaviour to cross-sectional dependence. The core historical universe contains 58 assets, producing $N(N - 1)/2 = 1653$ unique asset pairs. Figure 2 summarizes the full pairwise correlation distribution and then splits pairs into within-sector and between-sector groups.

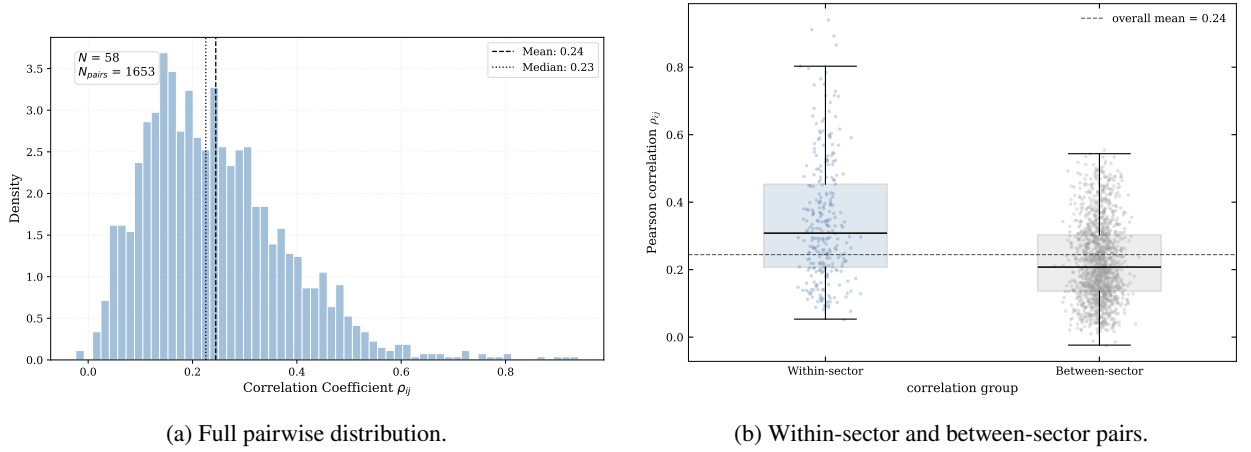


Figure 2: Correlation structure in the core historical universe. Panel (a) shows the full Pearson correlation distribution. Panel (b) compares within-sector and between-sector correlations using the initial macro-sector classification.

The full distribution is centred in a moderate positive range, with mean correlation close to 0.24 and median correlation close to 0.23. This level of average dependence is large enough to imply a strong common component, but the distribution remains dispersed. The dispersion is important because it leaves room for sectoral and subsectoral organization rather than a single homogeneous market block.

5.4. Sectoral correlation comparison

The sector split strengthens this interpretation. Within-sector correlations are visibly shifted to the right relative to between-sector correlations. This is the first direct evidence that the initial macro-sector map is reflected in the empirical dependency matrix.

Table 3

Within-sector and between-sector correlation summary.

Group	Pairs	Mean	Median	Std. dev.	Min	Max
Within-sector	275	0.3433	0.3082	0.1819	0.0534	0.9402
Between-sector	1378	0.2249	0.2076	0.1166	-0.0236	0.5552

The difference is economically meaningful. Within-sector pairs have mean correlation 0.3433, while between-sector pairs have mean correlation 0.2249. The one-sided Mann–Whitney test for the alternative that within-sector correlations are larger gives $p = 1.33389 \times 10^{-24}$. This does not make the sector map definitive, but it confirms that sectoral information is present in the correlation matrix and should be preserved in later network analysis.

5.5. Dynamic market synchronization

Static correlations are not enough because market dependence changes over time. Figure 3 shows the rolling average market correlation. This statistic compresses the correlation matrix into a time series that measures how synchronized the market is at each point in time.

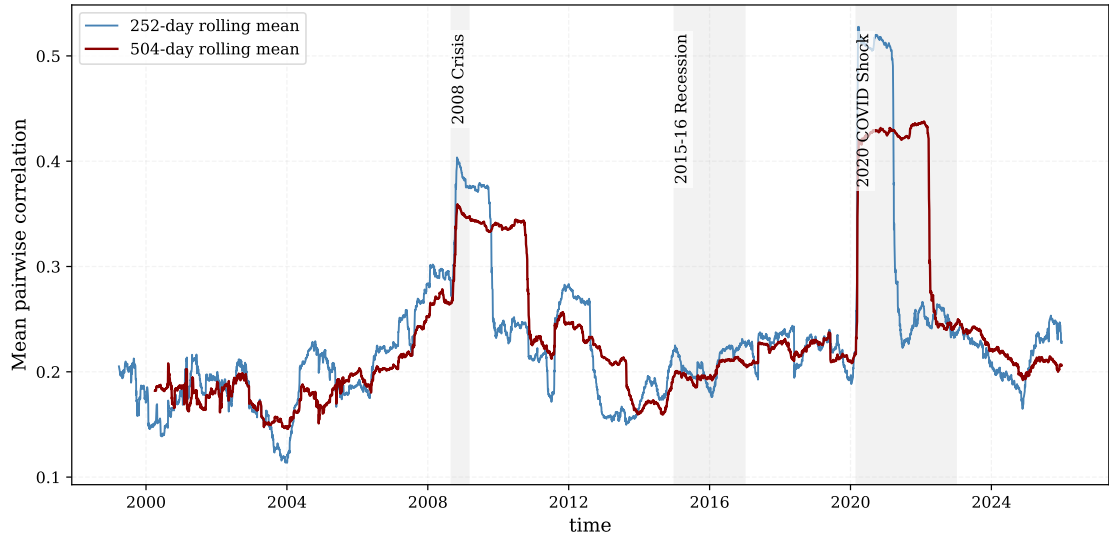


Figure 3: Rolling average market correlation for the core historical universe.

The rolling average correlation rises during stress periods, including the 2008 global financial crisis, the Brazilian recession period around 2015–2016 and the 2020 COVID shock. These episodes indicate stronger market-wide synchronization. This is precisely the kind of collective behaviour that can dominate the first eigenvalue of the correlation matrix.

Figure 4 adds a more granular view by showing selected pairwise dynamic correlations under different estimators. The objective is not to replace the full matrix but to show that individual dependencies can change substantially over time.

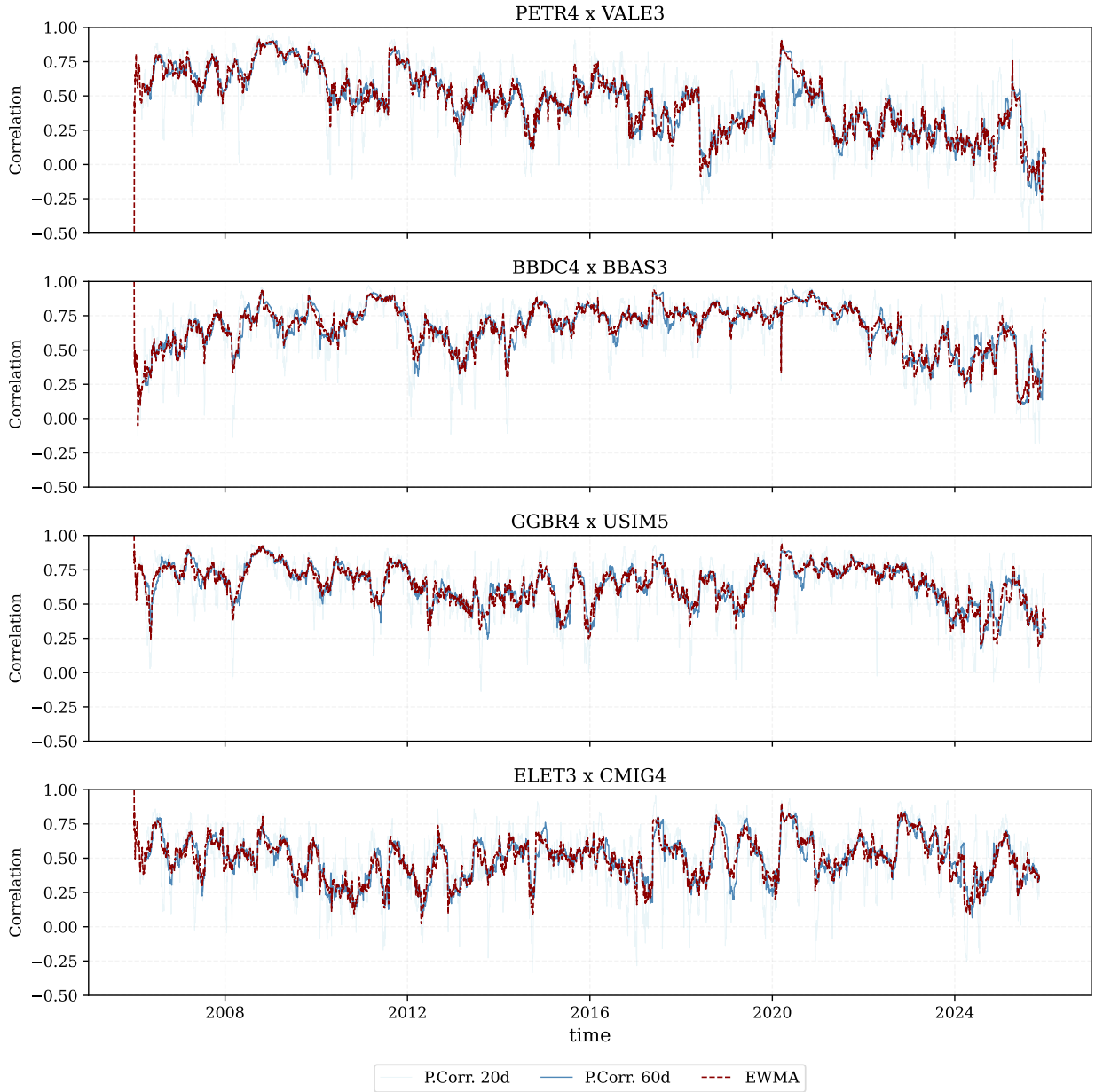


Figure 4: Dynamic pairwise correlations for selected asset pairs. Short-window, longer-window and EWMA estimates highlight the instability of individual dependencies.

The dynamic pairwise figure shows why a single full-sample correlation should be interpreted carefully. Short-window correlations react quickly but are noisy; longer-window estimates are smoother but slower; EWMA estimates provide an intermediate view. This motivates the use of RMT and filtered networks as structured summaries of a noisy and time-varying dependency system.

Table 4

RMT summary for the core historical universe.

N	T	Q	λ_-	λ_+	λ_1	λ_2	λ_3	λ_4	λ_5	Above λ_+
58	1527	26.33	0.6482	1.4278	21.6505	3.0957	1.7839	1.6543	1.4835	5

6. Random Matrix filtering

6.1. Empirical eigenspectrum

The RMT analysis is the bridge between raw correlations and filtered networks. It asks how many collective modes are distinguishable from random-matrix noise. The core historical matrix uses $N = 58$ assets and $T = 1527$ observations, giving $Q = T/N \approx 26.33$. The Marcenko–Pastur bounds are $\lambda_- = 0.6482$ and $\lambda_+ = 1.4278$.

Figure 5 shows the empirical spectrum against the random benchmark. The largest eigenvalue is far outside the Marcenko–Pastur band. Four additional eigenvalues also lie above the upper edge, suggesting that the correlation matrix contains a dominant market mode plus a small number of group or sector modes.

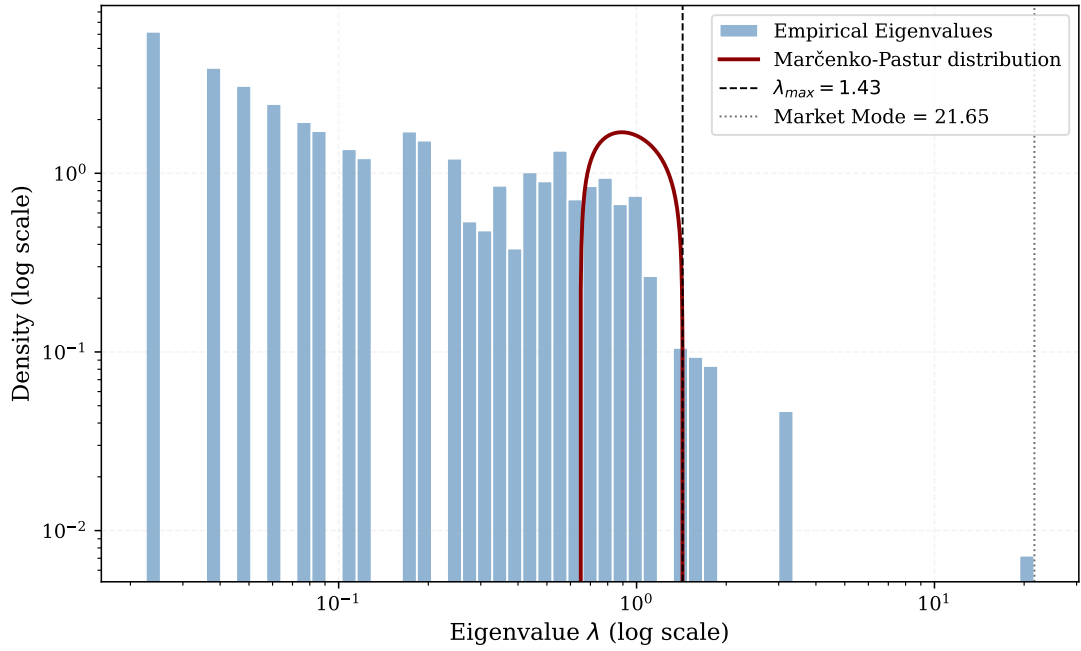


Figure 5: Empirical eigenvalue spectrum and Marcenko–Pastur bounds. The largest eigenvalue is far outside the random band and is interpreted as the Market Mode.

6.2. Market Mode and significant eigenvalues

The largest eigenvalue is 21.6505, corresponding to approximately 37.3% of the total trace of the correlation matrix. This is the Market Mode. It captures broad co-movement and explains why the original correlation matrix has a strong positive common component. The next four eigenvalues are 3.0957, 1.7839, 1.6543 and 1.4835. These values are above λ_+ and are interpreted as candidates for group or sector structure.

The key methodological point is that the market mode is informative but also visually dominant. If a network is built directly from the original matrix, many edges can reflect market-wide co-movement rather than local structure. The group-mode matrix is therefore constructed to ask what remains after the broad common component is removed.

6.3. Top eigenvector loadings

Eigenvalues identify the strength of collective modes, but eigenvectors help interpret their composition. Figure 6 reports top loadings for the leading eigenvectors. The first eigenvector is interpreted as the Market Mode. Subsequent eigenvectors are interpreted cautiously as group modes associated with sectors or localized dependency channels.

RMT-filtered networks for B3 volatility forecasting

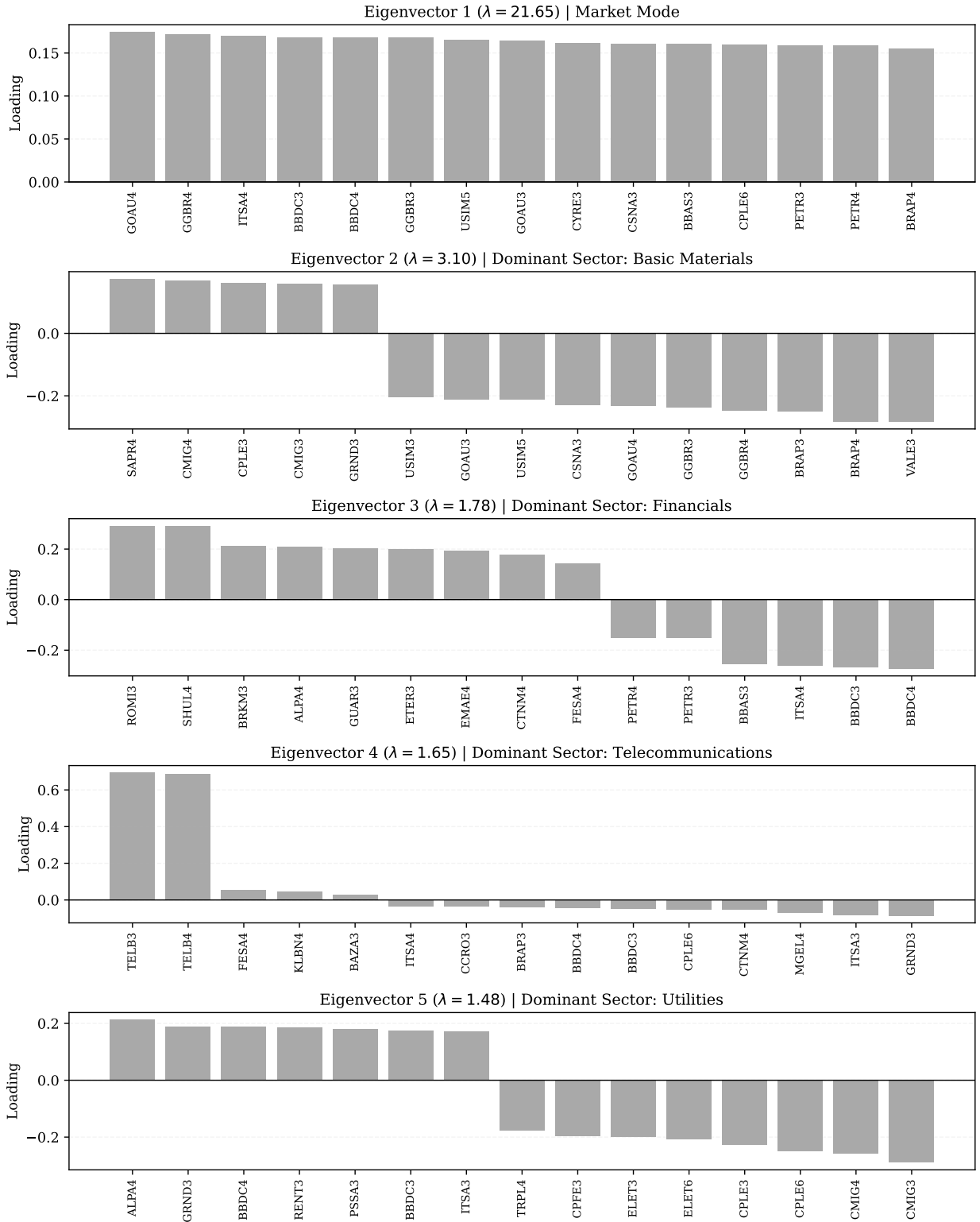


Figure 6: Top eigenvector loadings for the leading RMT modes. Eigenvector 1 corresponds to the Market Mode; subsequent eigenvectors are interpreted as group or localized sectoral components.

The second eigenvector is consistent with a Basic Materials and commodity-related component. The third contains financial and cross-sector structure. The fourth and fifth eigenvectors are more localized, with Telecommunications and Utilities-like components appearing among the dominant loadings. These labels are economic interpretations of the loading patterns. They should not be treated as mechanical classifications, because eigenvectors can mix sectors when firms share macroeconomic or liquidity exposures. The signs of eigenvector loadings are arbitrary up to multiplication by -1 ; therefore, interpretation focuses on relative grouping and magnitude rather than sign alone.

6.4. Matrix reconstruction

The spectral decomposition is numerically stable. Table 5 reports reconstruction diagnostics. The Frobenius reconstruction error is approximately 2.74×10^{-14} , and the maximum absolute reconstruction error is approximately 3.99×10^{-15} . The diagonal of the filtered matrix is exactly 1.0 for all assets, confirming that the spectral reconstruction does not introduce numerical artefacts. This check ensures that the subsequent network differences are caused by mode selection rather than numerical instability.

Table 5

RMT reconstruction diagnostics.

Diagnostic	Value
Frobenius reconstruction error	2.74×10^{-14}
Maximum absolute reconstruction error	3.99×10^{-15}
C_{filtered} diagonal min	1.0000
C_{filtered} diagonal max	1.0000
Eigenvalues above λ_+	5
Market-mode trace share	0.3733

Figure 7 shows the resulting matrix components. The original matrix combines all modes. The market-mode matrix captures broad positive co-movement. The group-mode matrix highlights weaker localized blocks after the market mode is removed. The noise component contains the random-band contribution, and the filtered matrix recombines the informative modes.

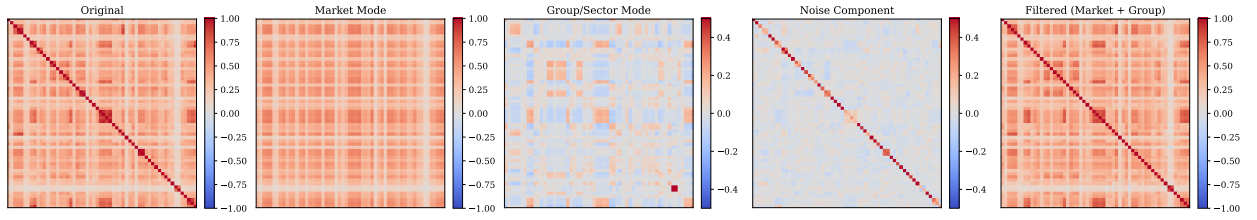


Figure 7: RMT-filtered matrix components. The decomposition separates original correlations into market, group/sector, noise and filtered components.

The important empirical result is not only that the market mode is large. It is that the group-mode matrix remains structured after removing it. This residual structure motivates the clustering and network sections: if localized dependency blocks remain, MST and PMFG topologies should change when built from the group-mode matrix.

7. Hierarchical clustering and ordered heatmaps

7.1. Ordered heatmaps

RMT separates the matrix into spectral components, but heatmaps help determine whether these components have visible block structure. Figure 8 orders the assets through hierarchical clustering and displays original, filtered and group-mode structures. The purpose is visual and diagnostic: the heatmaps show whether the spectral filtering produces economically interpretable blocks rather than only abstract eigenvalue differences.

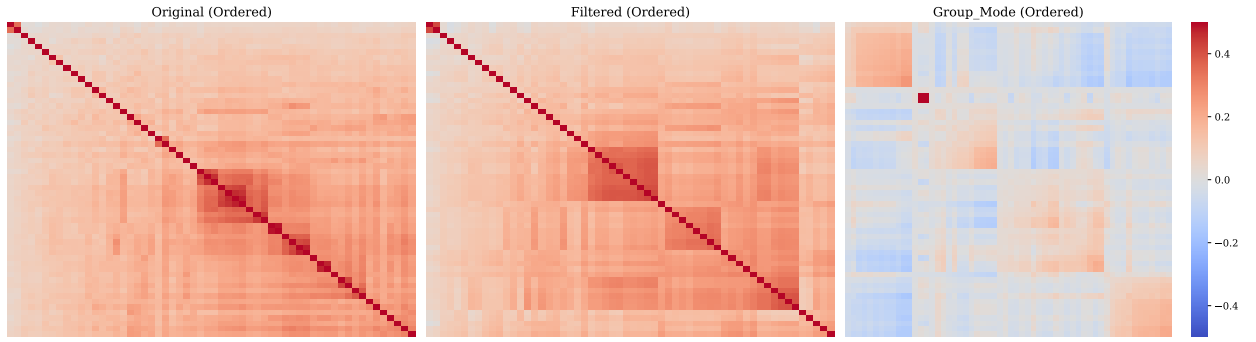


Figure 8: Ordered heatmaps for original and RMT-filtered correlation matrices. Hierarchical ordering reveals block structures that are less visible in unordered matrices.

The original matrix shows a broad positive background, consistent with the strong market mode. The filtered matrix preserves much of the organized dependency while suppressing noise. The group-mode heatmap is weaker in average correlation but more informative for localized blocks, because the dominant common component no longer masks sectoral variation. This is the visual counterpart of the RMT decomposition.

7.2. Dendrograms comparison

Dendrograms provide a complementary view of the same distance geometry. Figure 9 compares hierarchical clustering trees for original, filtered and group-mode matrices. These trees help identify which assets join early, which branches are stable and how the removal of the market mode changes the hierarchy.

RMT-filtered networks for B3 volatility forecasting

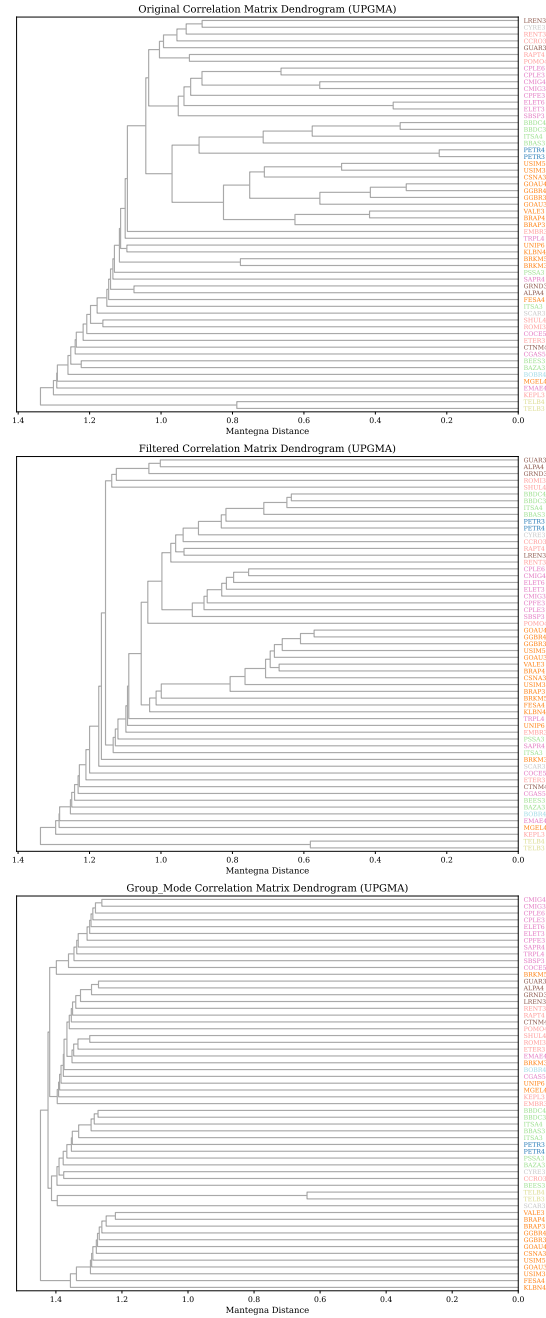


Figure 9: Dendrogram comparison for original, filtered and group-mode matrices using Mantegna distances and average linkage.

Table 6

Cophenetic correlations for hierarchical clustering.

Matrix	Assets	Cophenetic correlation
Original	58	0.9500
Filtered	58	0.9340
Group Mode	58	0.8705

The cophenetic correlation is highest for the original matrix, 0.9500, and remains high for the filtered matrix, 0.9340. The group-mode value is lower, 0.8705. This decline is expected rather than problematic. Removing the market mode reduces the smooth common structure that makes the hierarchy easier to summarize, leaving a more localized and less homogeneous dependency system.

The clustering results connect naturally to network filtering. Ordered heatmaps and dendrograms show that localized structure survives RMT filtering. The next section asks how that structure appears when only the strongest distance-ranked edges are retained under MST and PMFG constraints.

8. Financial network topology

Clustering summarises hierarchical organisation, but it does not identify graph-theoretic hubs, bridges or local cycles. Financial networks provide these complementary views by retaining only the strongest distance-ranked edges under structural constraints. We therefore move from dendrograms to filtered financial networks to study how dependency topology changes after RMT filtering.

8.1. MST backbone

The MST is the sparsest connected representation of the dependency geometry. With $N = 58$, it contains 57 edges. Figure 10 compares the MST built from the original matrix with the MST built from the group-mode matrix. This comparison is useful because it shows how much of the original backbone is driven by the market mode.

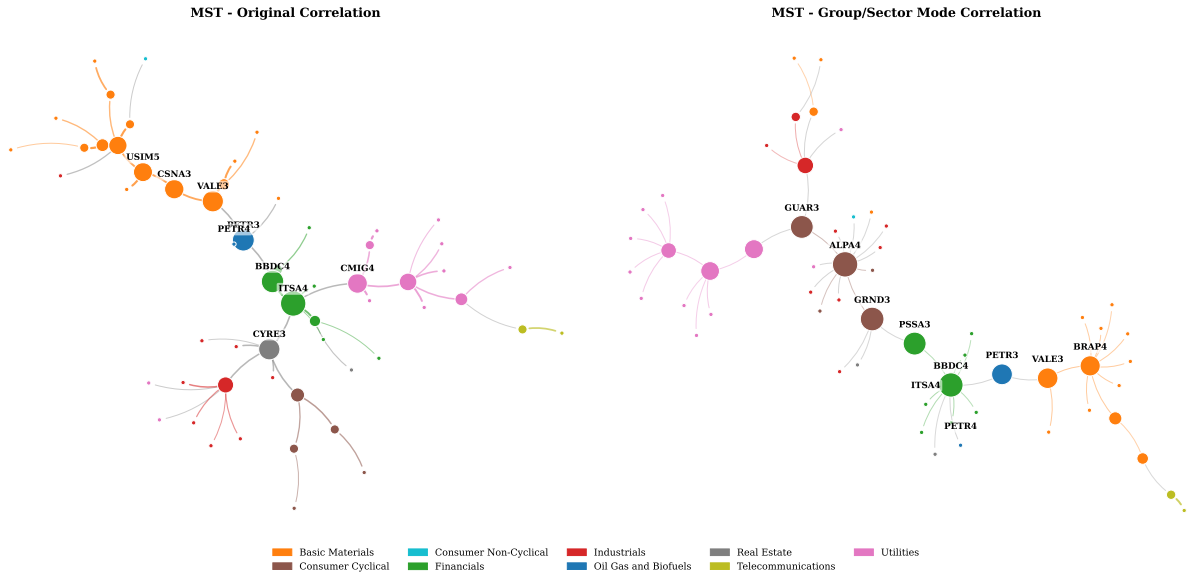


Figure 10: Refined MST comparison for original and group-mode correlation matrices. The original MST emphasizes strong market-mode dependencies, while the group-mode MST highlights weaker localized dependency channels.

The original MST has mean edge correlation 0.5796, while the group-mode MST has mean edge correlation 0.1388. This large decline is expected because the group-mode network excludes the dominant common component. The same-sector edge ratio also changes, from 0.7193 in the original MST to 0.6316 in the group-mode MST. The group-mode MST is therefore not a weaker version of the original tree; it is a different view of the market after broad synchronization is removed.

8.2. PMFG topology

The PMFG is richer than the MST because it preserves planarity while retaining $3N - 6 = 168$ edges. It therefore includes cycles, triangles and 4-cliques. Figure 11 compares original and group-mode PMFGs. This is the main network visualization in the paper because the PMFG retains enough structure to study local communities and hub reorganization.

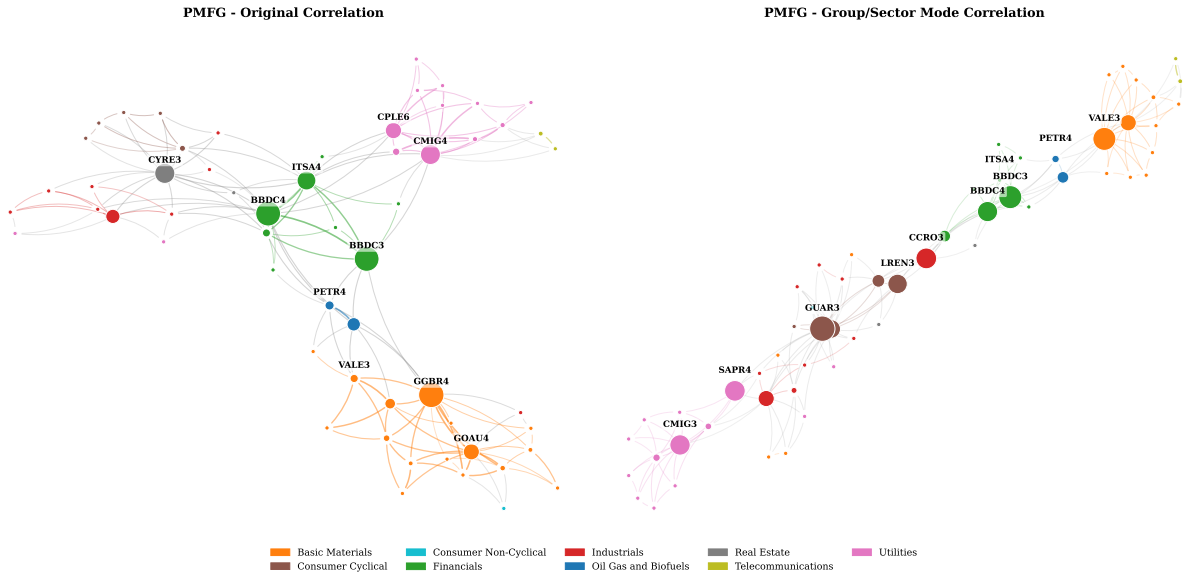


Figure 11: Refined PMFG comparison for original and group-mode correlation matrices. The group-mode PMFG reduces market-wide dominance and highlights more localized dependency channels.

The original PMFG has mean edge correlation 0.5177. The group-mode PMFG has mean edge correlation 0.1109. Despite this lower average edge strength, the group-mode PMFG has higher average clustering, 0.6653, than the original PMFG, 0.5273. This is a central result: removing the market mode lowers the absolute level of edge correlations, but it can reveal more clustered local structure.

8.3. Original vs Group/Sector Mode networks

Table 7 summarizes the main MST and PMFG statistics. It shows the same pattern across graph families. Original networks have stronger edges because they contain the market mode. Group-mode networks have weaker edges but preserve meaningful sectoral and subsectoral organization.

Table 7

Selected MST and PMFG topology statistics.

Network	Matrix	Edges	Mean edge corr.	Same-sector ratio	Clustering	Top degree	Top betweenness	4-cliques
MST	Original	57	0.5796	0.7193	0.0000	RAPT4	ITSA4	0
MST	Group mode	57	0.1388	0.6316	0.0000	ALPA4	ALPA4	0
PMFG	Original	168	0.5177	0.6012	0.5273	BBDC4	GGBR4	55
PMFG	Group mode	168	0.1109	0.5417	0.6653	GUAR3	GUAR3	55

8.4. Network topology metrics

Figure 12 expands the comparison beyond average edge correlation. It summarizes mean edge correlation, same-sector ratio, same-subsector ratio and average clustering across network and matrix types. This figure is useful because it shows that topology is not reducible to a single edge-weight statistic.

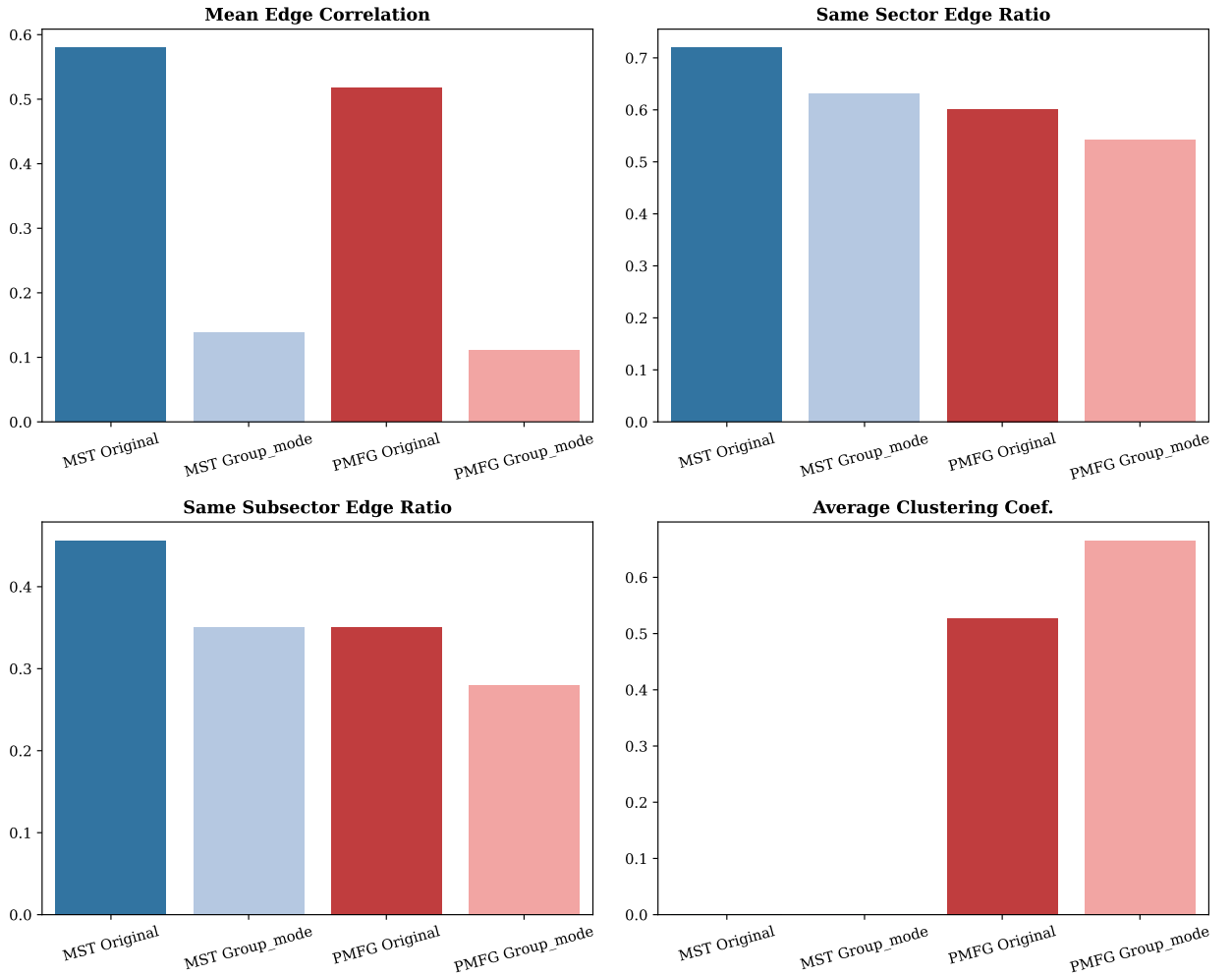
Topology Metrics Comparison: MST vs PMFG

Figure 12: Network topology comparison across original, filtered and group-mode MST/PMFG representations.

The topology comparison supports a mesoscopic interpretation. Group-mode networks have lower mean edge correlations, but the PMFG clustering coefficient increases. Same-sector and same-subsector ratios remain substantial even after market-mode removal. This pattern is consistent with localized dependency structure rather than pure random residual noise. MST clustering is zero by construction because trees do not contain triangles; the PMFG clustering coefficient is therefore a distinctive advantage of the richer planar representation.

Table 8

PMFG clique and planarity diagnostics.

Matrix	Nodes	Edges	Triangles	4-cliques	Planar
Original	58	168	166	55	Yes
Filtered	58	168	166	55	Yes
Group Mode	58	168	166	55	Yes

The clique counts are also construction checks. For a PMFG with 58 nodes, 168 edges are expected. The presence of 166 triangles and 55 4-cliques confirms that the PMFG contains richer local geometry than the MST. These local structures are precisely what can be transformed into network predictors for the forecasting experiment.

8.5. Hub reorganization after RMT filtering

Hub rankings reveal which assets act as bridges or local centers. Figure 13 compares hub rankings before and after filtering. In the original PMFG, GGBR4 is the top betweenness node and BBDC4 is the top degree node. In the group-mode PMFG, GUAR3 becomes the leading node by both degree and betweenness.

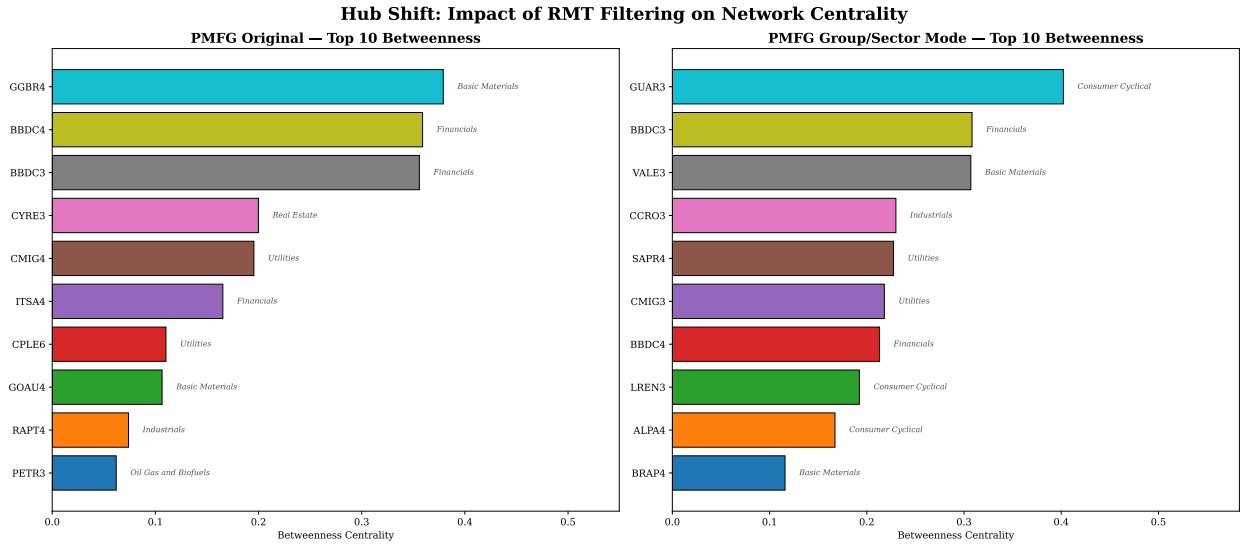


Figure 13: Hub-rank comparison across original and group-mode networks. RMT filtering changes which assets act as central bridges in the dependency topology.

This hub shift is important. Original hubs can reflect market-wide correlation strength, liquidity and broad exposure. Group-mode hubs are more likely to represent localized bridges after the common component is removed. The appearance of assets such as GUAR3, BBDC3, VALE3, CCRO3, SAPR4 and CMIG3 in group-mode rankings suggests that RMT filtering changes the economic interpretation of centrality.

9. Aggregated subsector dependency network

Asset-level networks can be difficult to interpret when many share classes and sectors are present. The aggregated subsector dependency network converts ticker-level dependencies into a mesoscopic map. Nodes represent subsectors and edges summarize average dependency strength between subsectors. This is analogous to sector or country aggregation maps used in the financial dependency literature, but adapted to the Brazilian equity universe.

Table 9

Aggregated subsector network summary.

Subsectors	Edges	Density	Mean edge dep.	Median edge dep.	Top degree	Top betweenness	Top weighted degree
22	66	0.2857	0.0566	0.0561	Retail	Electric Utility	Apparel and Footwear

Figure 14 presents the resulting network. The network contains 22 subsectors and 66 edges, with density 0.2857. The mean edge dependency is 0.0566 and the median edge dependency is 0.0561. These values are lower than raw asset-level correlations because the network is built after aggregation and filtering, but the topology remains interpretable.

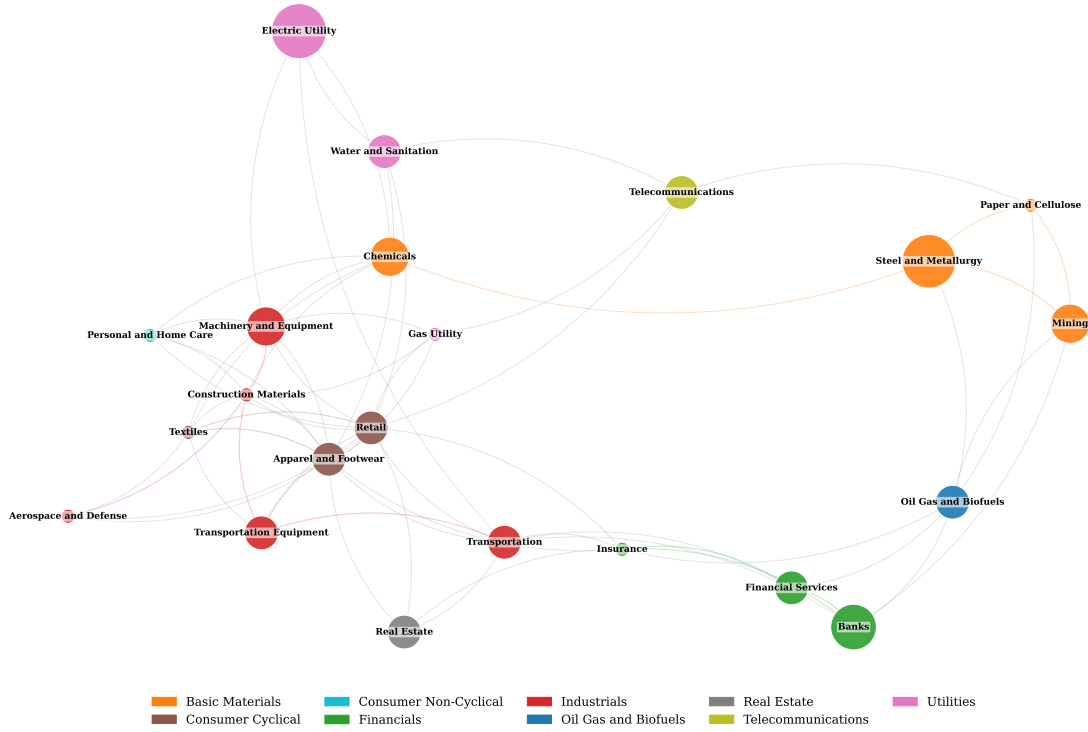
B3 Subsector Aggregated Dependency Network

Figure 14: Aggregated subsector dependency network. Asset-level dependencies are summarized into subsector-level connections, providing a mesoscopic view of the Brazilian equity market.

The top degree subsector is Retail, indicating broad direct connectivity in the aggregated dependency map. Electric Utility has the highest betweenness, suggesting that it acts as a bridge in the mesoscopic topology. Apparel and Footwear has the highest weighted degree, reflecting strong dependency intensity across its retained connections. These summaries help translate the abstract PMFG topology into economic language.

The subsector network also provides a useful bridge to forecasting. If network topology contains information beyond univariate volatility persistence, then node-level or group-level centrality measures may help explain future volatility regimes. The forecasting section tests this idea in a conservative first experiment.

10. Volatility forecasting

10.1. Forecasting dataset and temporal split

The forecasting experiment evaluates whether RMT and network features contain incremental information for realized volatility. The experiment is deliberately conservative. It uses PETR4, VALE3 and BBDC4 as target assets and compares network-augmented machine-learning models with strong econometric baselines. The data are split temporally: training covers 2006–2017, validation covers 2018–2020 and testing covers 2021–2025. This strict temporal split avoids random cross-sectional leakage across time and ensures that model selection is not based on future observations.

10.2. GARCH persistence and econometric benchmarks

The GARCH(1,1) estimates with Student- t innovations confirm high volatility persistence. Table 10 reports ω , α , β and $\alpha + \beta$. The persistence values are 0.9911 for PETR4, 0.9946 for VALE3 and 0.9828 for BBDC4. These high values explain why classical volatility benchmarks are difficult to outperform and are consistent with the slow ACF decay of absolute returns documented in Section 5.

Table 10

GARCH(1,1) persistence estimates for the forecasting assets.

Asset	ω	α	β	$\alpha + \beta$
BBDC4	0.0729	0.0520	0.9308	0.9828
PETR4	0.0780	0.0758	0.9153	0.9911
VALE3	0.0417	0.0512	0.9434	0.9946

10.3. Machine-learning feature sets

The machine-learning models use three nested feature sets in an ablation design. Set A contains classical volatility features. Set B adds market and RMT variables. Set C adds MST and PMFG network variables. This nesting allows the paper to isolate whether network topology adds information beyond classical lagged-volatility persistence and broad market-mode features.

10.4. Out-of-sample model comparison

Figure 15 summarizes out-of-sample QLIKE loss. Lower QLIKE is better. The figure uses the best feature set for each machine-learning model and horizon, avoiding ambiguous labels and making the comparison easier to interpret.

RMT-filtered networks for B3 volatility forecasting

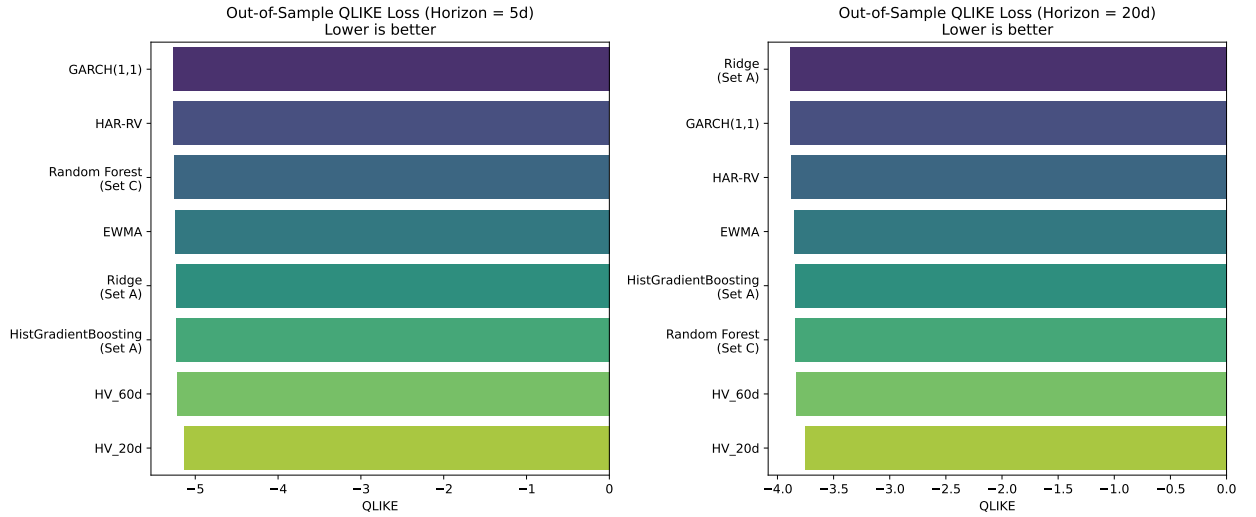


Figure 15: Volatility forecasting comparison for econometric and machine-learning models. Machine-learning labels show the best feature set selected for each model and horizon.

At the 5-day horizon, GARCH(1,1), HAR-RV and Random Forest are highly competitive. GARCH has average test QLIKE close to -5.27, HAR-RV is close to -5.26 and Random Forest with Set C is close to -5.25. The differences are small, which means the safe interpretation is not that machine learning dominates. Rather, the network-augmented model approaches strong classical benchmarks.

At the 20-day horizon, Ridge models with Set A or Set B compete strongly and slightly outperform GARCH in average QLIKE. Ridge Set A and Set B are close to -3.89, while GARCH is close to -3.88. Random Forest Set C remains competitive but does not dominate. This result suggests that longer-horizon volatility may be easier to forecast with stable linear shrinkage features, while nonlinear models require careful regularization and validation.

Table 11

Aggregated test-set forecasting comparison (2021–2025). Values are averaged across PETR4, VALE3 and BBDC4.

Horizon	Model	Feature set	MAE	RMSE	QLIKE	R^2_{oos}
5	GARCH(1,1)	Classical	0.0156	0.0213	−5.27	0.16
5	HAR-RV	Classical	0.0148	0.0198	−5.26	0.22
5	Random Forest	Set C (+Network)	0.0151	0.0204	−5.25	0.20
5	EWMA	Classical	0.0153	0.0207	−5.25	0.19
20	Ridge	Set A (Classical)	0.0241	0.0328	−3.89	0.39
20	Ridge	Set B (+Market/RMT)	0.0242	0.0329	−3.89	0.39
20	GARCH(1,1)	Classical	0.0278	0.0381	−3.88	0.23
20	HAR-RV	Classical	0.0260	0.0356	−3.88	0.28
20	Random Forest	Set C (+Network)	0.0287	0.0393	−3.85	0.17

10.5. Realized vs predicted volatility diagnostics

Aggregate metrics can hide important time-series behaviour. Figure 16 compares realized 20-day volatility with GARCH and Random Forest forecasts for PETR4, VALE3 and BBDC4. This diagnostic is included in the body because it shows how models react to volatility regimes.

RMT-filtered networks for B3 volatility forecasting

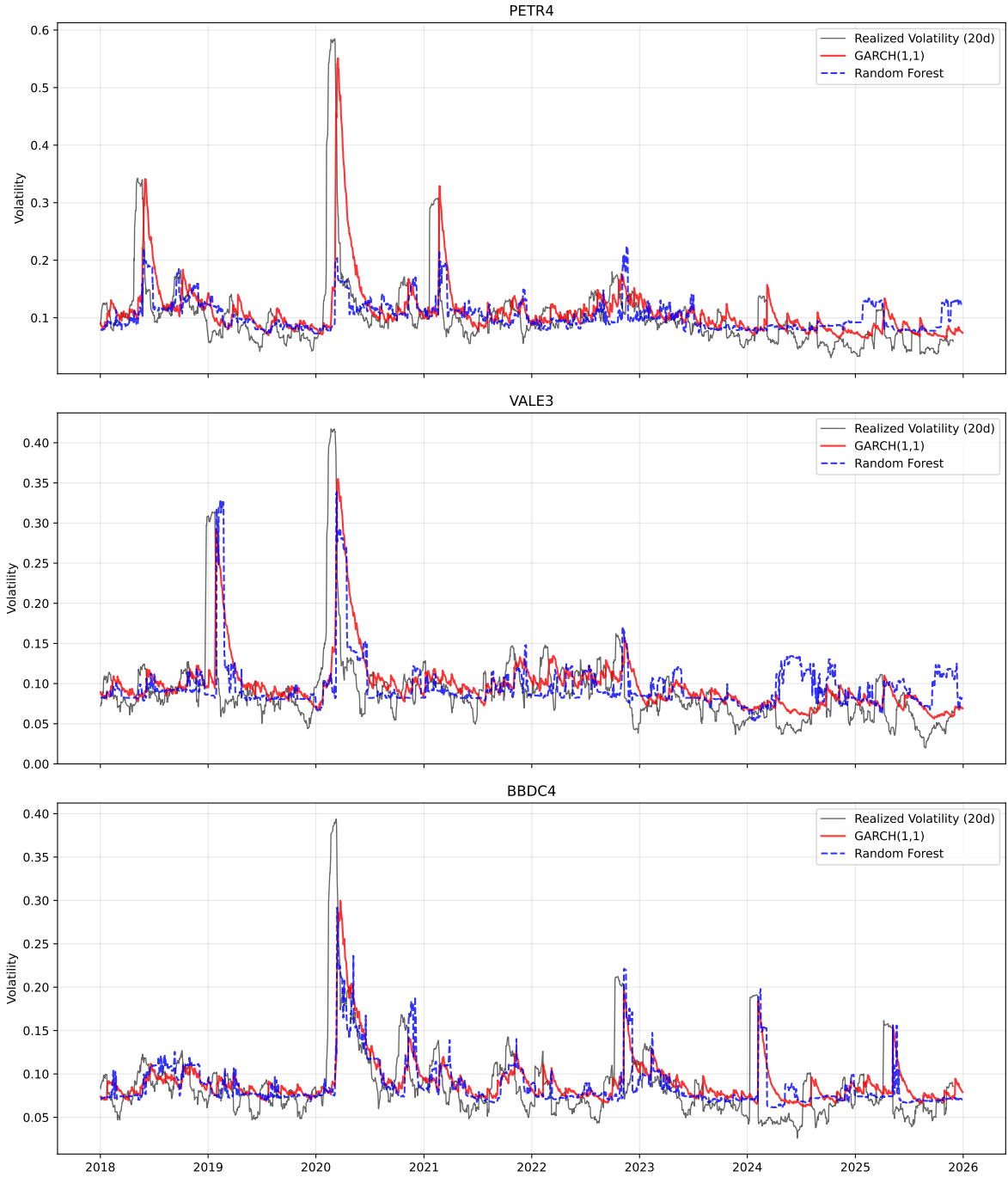


Figure 16: Realized versus predicted 20-day volatility for selected models and assets. The plot illustrates regime tracking and forecast smoothness.

The diagnostic plots show that broad volatility regimes are captured, but extreme spikes remain difficult. GARCH reacts strongly to shock persistence, while Random Forest predictions can be smoother or more step-like depending on feature regimes. This behaviour is consistent with the aggregate results: nonlinear models can be competitive, but the dominant forecasting signal remains volatility persistence.

RMT-filtered networks for B3 volatility forecasting

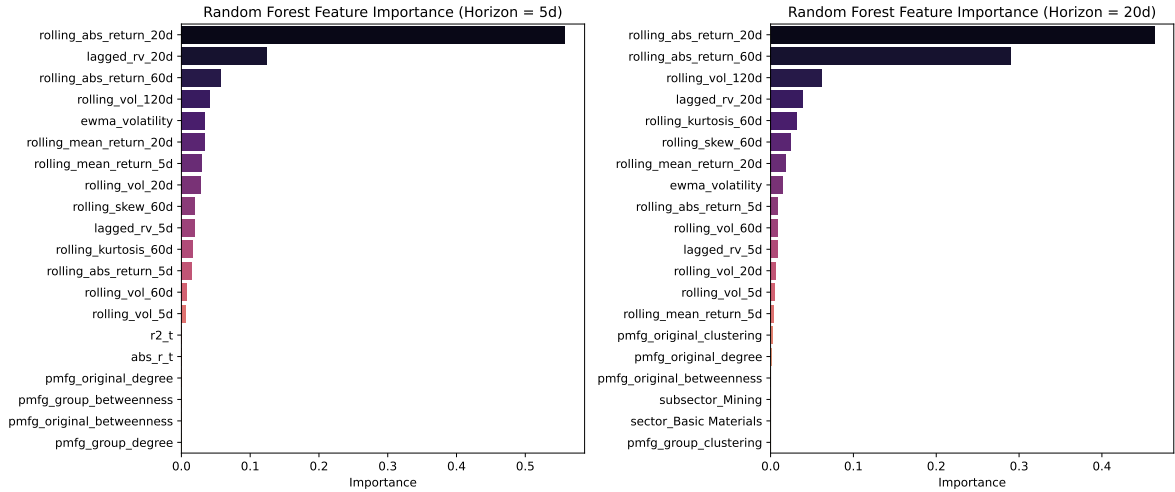


Figure 17: Feature importance for Random Forest volatility forecasting models. Classical volatility features dominate, while PMFG and MST variables have smaller but nonzero importance in the network-augmented feature set.

10.6. Feature importance and network signal

Feature importance provides a direct view of what the Random Forest uses. Figure 17 shows that classical volatility features dominate the model. Rolling absolute returns, lagged realized volatility, rolling volatility and EWMA variables carry most of the importance.

The strongest features are rolling absolute return over 20 days, lagged realized volatility and longer rolling volatility measures. Network features are much smaller, but they are not zero. PMFG clustering, PMFG degree and PMFG betweenness appear among the nonzero network predictors, especially at the 20-day horizon. The appropriate conclusion is therefore incremental: topology does not replace classical volatility features, but it may add structural information when included in a broader feature set.

11. Discussion

Taken together, the empirical results support a layered interpretation of Brazilian equity dependencies. The first layer is the Market Mode. It is visible in the moderate positive average correlation ($\bar{\rho} \approx 0.24$), the increase in rolling average correlation during stress periods and the very large first eigenvalue ($\lambda_1 = 21.65$, approximately 37.3% of the spectrum). This mode is economically meaningful because it captures broad market synchronization driven by domestic monetary policy, exchange-rate dynamics and global commodity cycles. It is also methodologically important because it can dominate raw networks and obscure weaker local structures.

The second layer is group and sector structure. Within-sector correlations are statistically significantly higher than between-sector correlations ($p \approx 1.33 \times 10^{-24}$), and the RMT decomposition identifies four additional eigenvalues above the random upper edge. Eigenvector loadings and group-mode matrices suggest that commodity, financial, telecommunications and utility-related structures contribute to the non-random part of the spectrum. These structures are not as strong as the market mode, but they are precisely the level where network filtering becomes informative.

The third layer is noisy residual dependence. Not every correlation is meaningful, and not every edge retained by a graph should be interpreted economically. This is why the paper combines RMT, heatmaps, dendrograms and network metrics. A result is more credible when it appears consistently across several representations rather than only in one visualization.

The network results demonstrate that market-mode removal changes topology in a structured way. Original networks have stronger average edge correlations, while group-mode networks have much lower average edge correlations. However, the group-mode PMFG has higher clustering (0.6653 versus 0.5273). This combination is a central finding: lower average edge strength does not mean absence of structure. It means that the dominant common component has been removed and localized mesoscopic organization becomes easier to observe. The PMFG is particularly suited for this analysis because it preserves richer topology than the MST, retaining 168 edges with 166

triangles and 55 4-cliques that allow hub-rank shifts, clustering changes and subsector dependency maps to be studied in detail.

Clustering summarizes hierarchical organization, but it does not identify graph-theoretic hubs, bridges or local cycles. The transition from dendrograms to filtered financial networks reveals complementary information. The hub reorganization after RMT filtering is particularly striking: original hubs such as GGBR4 and BBDC4 reflect market-wide correlation strength and broad exposure, while group-mode hubs such as GUAR3 represent localized bridges after the common component is removed. This suggests that centrality in financial networks has different economic interpretations depending on whether the market mode is included or excluded.

The forecasting results should be interpreted with appropriate caution. They do not show that machine learning uniformly dominates classical volatility models. Instead, they show that network-augmented models can approach strong baselines and that PMFG and MST variables have nonzero importance in Random Forest models. The persistence of GARCH(1,1), with $\alpha + \beta$ consistently above 0.98, confirms that autoregressive volatility dynamics remain the dominant forecasting signal for Brazilian equities. Network features complement rather than replace this signal. The appropriate interpretation is incremental: topology adds structural information from the cross-sectional geometry of the market to the temporal dynamics of individual asset variance.

The broader contribution is methodological. The paper connects econophysics filtering, financial networks and volatility forecasting in a single reproducible pipeline for B3 equities. By documenting each step transparently and evaluating network features against demanding econometric benchmarks, the analysis provides a template that can be extended to other emerging markets, different time scales and more sophisticated forecasting architectures.

12. Limitations

The first limitation concerns time variation. The RMT and network features used in the initial forecasting exercise are based on full-sample or slowly changing structures. They are useful for understanding the market, but they should not be interpreted as fully real-time predictors unless reconstructed recursively using only information available at each forecast origin. A stricter forecasting design should use rolling RMT and rolling network features.

The second limitation concerns the target set. The forecasting experiment uses PETR4, VALE3 and BBDC4. These assets are liquid and economically important, but they do not cover the full B3 cross-section. The next version should extend the forecasting exercise to a larger set of assets and evaluate whether network features help more for some sectors than others.

The third limitation concerns data quality. The core historical universe is sufficiently clean for the current analysis, but the modern liquid top-50 universe contains extreme adjusted-return events that require additional validation. These events may reflect corporate actions, ticker transitions, units or adjustment-factor issues. They should be flagged, inspected and either corrected or handled through robustness checks before the modern universe is used for final network claims.

The fourth limitation concerns sector classification. The macro-sector and subsector map is an initial classification. It is adequate for testing whether within-sector correlations exceed between-sector correlations, but it should be reviewed ticker by ticker before submission. This is especially important for holding companies, units, firms with multiple share classes and companies with mixed exposure across sectors.

Finally, the machine-learning results are preliminary. Feature importance in tree models is useful but not a causal explanation. The nonzero importance of PMFG features suggests incremental signal, but the magnitude is small compared with classical volatility predictors. Future work should include rolling network features, DCC-GARCH comparisons, more target assets, temporal neural architectures and graph neural networks.

13. Conclusion

This paper develops a comprehensive econophysics pipeline for analysing the dependency structure of Brazilian equities traded on B3 from 1998 to 2025. The main findings can be summarised along four dimensions.

First, the stylized-fact and correlation analysis confirms that Brazilian equities display the empirical signatures expected from complex financial systems: fat-tailed return distributions, persistent volatility clustering and a moderate but positive average cross-correlation ($\bar{\rho} \approx 0.24$). Within-sector correlations are statistically significantly higher than between-sector correlations, providing early evidence that sectoral organisation is embedded in the empirical dependency matrix.

Second, Random Matrix Theory separates the correlation matrix into interpretable components. Five eigenvalues exceed the Marčenko–Pastur upper bound, with the largest eigenvalue accounting for approximately 37.3% of the total variance. The Market Mode captures broad co-movement, while the remaining significant eigenvectors reveal commodity, financial and utility-related group structures. The spectral reconstruction is numerically stable, and the group-mode matrix retains meaningful localized dependency blocks after the dominant market component is removed.

Third, financial network analysis through MST and PMFG representations shows that RMT filtering changes the topology of asset dependencies. The group-mode PMFG has lower mean edge correlation but higher clustering than the original PMFG, indicating that market-mode removal reveals denser localized communities. Hub rankings shift from market-wide connectors (GGBR4, BBDC4) to localized sectoral bridges (GUAR3). The PMFG retains 168 edges with 166 triangles and 55 four-cliques, providing richer local geometry than the MST and enabling more detailed mesoscopic analysis.

Fourth, the volatility forecasting experiment provides evidence that network topology can complement classical volatility predictors. GARCH(1,1) and HAR-RV remain strong benchmarks, especially at short horizons, consistent with the high persistence ($\alpha + \beta > 0.98$) observed in all three target assets. Machine-learning models with PMFG-derived features approach these baselines and, at the 20-day horizon, Ridge Regression slightly outperforms GARCH in QLIKE. Feature importance analysis shows that while classical autoregressive features dominate, PMFG centralities contribute nonzero predictive information. The appropriate conclusion is incremental: network features do not replace classical volatility dynamics, but they add structural information from the cross-sectional geometry of the market.

Future work should extend this framework in several directions: constructing rolling RMT and network features for fully real-time forecasting, expanding the target asset set beyond the three demonstration equities, comparing against multivariate volatility models such as DCC-GARCH, and evaluating temporal neural network and graph neural network architectures that can jointly exploit the spatial structure of financial networks and the temporal dynamics of volatility. The broader goal is to connect interpretable complex-systems descriptions of financial dependency with practical risk forecasting in emerging markets.

A. Supplementary figures

This appendix is reserved for supplementary figures. In the current version, the main empirical figures are retained in the body to support the full research narrative. Figures that may be relocated here in a shortened submission include the complete eigenvector loading panel, the original MST and PMFG rough-comparison visualisations and additional dynamic pairwise correlation diagnostics.

B. Supplementary tables

This appendix is reserved for supplementary tables. The reproducibility materials generated by the computational pipeline include asset lists with sector and subsector labels, return summaries, correlation matrices, pairwise observation counts, RMT eigenvector loadings, network edge lists, clique lists, centrality rankings and the full forecasting dataset. A compact asset-universe table and a detailed forecasting comparison table by asset, horizon, model and feature set will be included here in the final submission version.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The raw market data used in this study were obtained from B3 daily price records processed through the QuantBase data infrastructure. Due to data redistribution restrictions, raw data are not redistributed with this manuscript. Derived tables, scripts and reproducibility materials will be made available in a public repository upon publication.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT for language refinement, structural organization and code-review support. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRedit authorship contribution statement

Andrey V. S. Souza: Conceptualization, Methodology, Software, Formal analysis, Writing - original draft. **Bruno L. Dalmazo:** Methodology, Supervision, Writing - review and editing. **Giancarlo Lucca:** Methodology, Validation, Writing - review and editing. **Eduardo N. Borges:** Supervision, Methodology, Writing - review and editing. **Richard F. Pinto:** Software, Data curation, Validation. **Fabian C. Cardoso:** Data curation, Resources, Writing - review and editing. **Viviane L. D. de Mattos:** Formal analysis, Supervision, Writing - review and editing. **Rafael A. Berri:** Supervision, Project administration, Writing - review and editing.

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